

POC: AI-Powered Real Estate Lead Qualification & Appointment Automation

1. Objective

Demonstrate an AI conversational agent that automates the early real estate lead journey: Lead Capture → AI Conversation → Qualification → Scoring → Appointment Booking → CRM Update → AI Summary → Sales Handover — reducing manual effort on the sales team and proving the core workflow before a larger build.

2. Business Problem

Leads arrive from multiple channels (portals, ads, social, website forms) and sales teams spend significant time on repetitive first-contact work: qualifying, scheduling, and CRM data entry. This POC automates that first stage.

3. Scope

In Scope

- One live lead source (website form) + simulated portal/ad/social leads via webhook
- AI conversational agent for natural-language qualification (no menu-driven chatbot)
- Configurable qualification framework with automated lead scoring (Qualified / Partially Qualified / Not Qualified)
- Appointment booking with one calendar integration (Google Calendar / Calendly)
- CRM sync (mock CRM or HubSpot)
- AI-generated lead summary and automated sales-agent assignment
- Small property knowledge base (RAG) to answer basic prospect questions
- Admin dashboard, lead details page, conversation history

4. Example Flow

Lead submits an enquiry → AI agent opens a natural conversation, asks about buyer type, purpose, budget, location, property type, financing, and timeline → system scores the lead (e.g. 85/100 – Highly Qualified) → AI offers appointment slots → prospect books → CRM record created/updated → AI summary generated → lead auto-assigned to the right sales rep with full context.

5. Qualification Parameters

Parameter	Example
Buyer Type	Investor
Purpose	Investment
Location	Dubai
Property Type	Apartment

Parameter	Example
Budget	\$500,000
Financing	Mortgage
Timeline	3–6 months

6. Technology Stack

Layer	Choice
Frontend	React / Next.js — chat UI + admin dashboard
Backend	Python, FastAPI, REST + webhooks
AI	OpenAI-compatible LLM, prompt-based qualification engine, RAG knowledge base
Database	PostgreSQL
Automation	n8n
Calendar	Google Calendar API / Calendly
CRM	Mock CRM or HubSpot
Hosting	Vercel (frontend), Render/AWS (backend), managed Postgres

7. Build Timeline — 4 Weeks

Week	Focus	Key Deliverables
Week 1	Foundation	Requirements & qualification framework finalized; architecture, DB, n8n set up; lead capture API; sample knowledge base
Week 2	AI Agent	Conversational AI, qualification workflow, conversation memory, knowledge-base integration, data extraction, lead scoring
Week 3	Automation & Integrations	Appointment scheduling, calendar integration, CRM sync, lead assignment, AI summary generation, sales notifications
Week 4	Dashboard & Demo	Admin dashboard, lead details page, conversation history, UI polish, end-to-end testing, deployment, client demo

8. Team

Role	Responsibility
AI/Backend Developer	AI agent, APIs, business logic
Full-Stack Developer	Conversational UI, dashboard, integrations
Automation Developer	n8n workflows, third-party integrations
QA Engineer	Functional and integration testing
Project/Technical Lead	Architecture, requirements, coordination

9. Acceptance Criteria

- A test lead can be submitted and the AI engages it automatically
- AI holds a natural conversation and extracts qualification data from free text
- System scores and classifies the lead, and qualified leads can book an appointment
- Appointment and lead data sync automatically to the CRM
- AI summary is generated and lead is auto-assigned to a sales rep
- Full workflow runs end-to-end with no manual data entry

10. Success Metrics

KPI	Target
Lead Response Time	Near real-time
AI Engagement Rate	> 90% of test leads
Qualification Completion	> 80%
CRM Sync Success	100% of test cases
Human Intervention	Minimal

11. Next Steps

Following a successful POC, future phases can extend the platform to WhatsApp/voice channels, multi-CRM support, advanced scoring and nurturing, and eventually a multi-tenant, multi-industry SaaS offering — while reusing the same core architecture.